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Investigating the responsibilities of digital organizations in facing sustainability challenges: Developing an empirical framework and measurement model for corporate digital responsibility

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ABSTRACT

The widespread adoption of digital technologies by organizations has led to the emergence of significant challenges for themselves and society. Hence, Corporate Digital Responsibility has been introduced as a solution for the responsible use of technologies that meet the expectations of stakeholders. This research aims to develop a validated scale for measuring Corporate Digital Responsibility and to examine its impact on digital sustainability in organizations. In the qualitative phase, three key dimensions—ethical, legal, and sustainability-oriented responsibilities—along with 11 indicators were identified through a combination of a literature review and Delphi interviews. In the quantitative phase, exploratory and confirmatory factor analyses were conducted on survey data from organizational representatives, confirming the robustness of the proposed measurement model. The findings confirm the proposed measurement model for Corporate Digital Responsibility and demonstrate that Corporate Digital Responsibility practices significantly enhance digital sustainability by promoting transparency, accountability, and environmental responsibility. These results provide a structured framework for researchers and actionable recommendations for managers to implement Corporate Digital Responsibility and effectively address the challenges of digital transformation.

1. Introduction

The pervasiveness of digital technologies is inducing significant transformations within both the personal and professional spheres of human existence (Gill et al., 2024). This phenomenon fosters the emergence of novel business paradigms and facilitates the adoption of innovative lifestyles. Consequently, a multitude of individuals and organizations have actively pursued digital transformation initiatives (Cardinali & De Giovanni, 2022). However, digital transformation, like any major disruption, creates both opportunities and challenges for organizations and society (Grybauskas et al., 2022; Švarc et al., 2024). Although digital technologies offer many benefits for disadvantaged social groups, the persistence of unequal access to the digital landscape presents a significant challenge (Bailey & Ngwenyama, 2011). In addition, digital transformation has negative consequences, such as unemployment due to automation and robotics or the risk of data breaches and cyber-attacks (Al-Khateeb et al., 2017). Consequently, the proliferation of digital technologies necessitates the acknowledgment of novel social issues and the assumption of heightened responsibilities, particularly for organizational entities (Lobschat et al., 2021). The imperative for the ethical deployment of digital technologies and data necessitates the establishment, within organizational cultures, of a comprehensive and cohesive framework of ethical norms. This framework would serve to govern the development and utilization of these technologies and data (Mihale-Wilson et al., 2022). In addition, the complex interplay between technological advancements and data utilization presents organizations with a multitude of ethical dilemmas. To navigate this evolving landscape effectively, organizations must cultivate a deeper understanding of ethical frameworks and foster a culture of responsible action (Lobschat et al., 2021). Although corporate responsibilities, especially corporate social responsibility, are well-recognized and leveraged by organizations, they have not been able to address the challenges and threats associated with the emergence of digital technologies (Herden et al., 2021; Lobschat et al., 2021; Mihale-Wilson et al., 2022; Mueller, 2022). Digitalization transcends a mere linear progression from past technological advancements, constituting a paradigm shift

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0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

characterized by exponential growth and fundamental changes in how we interact with and leverage technology (Lobschat et al., 2021). The heterogeneous nature of emerging technologies, exemplified by artificial intelligence's impact on value creation and societal interactions, entails a separate approach. This approach would involve the independent identification of novel social responsibilities alongside the development and implementation of such technologies.

One potential mitigation strategy for the challenges posed by novel digital technologies lies in the implementation of organizational policies that foster responsible approaches to their development, utilization, and continuous improvement. While such policies should be considered as part of a broader corporate responsibility, in recent years, researchers proposed the concept of corporate digital responsibility (CDR) (Mueller, 2022). Corporate Digital Responsibility has been proposed as a framework for managing ethical, privacy, and equity risks in the use of digital technologies in service companies (Basu et al., 2025; Tóth & Blut, 2024). It empowers organizational leadership to cultivate responsible application of emerging technologies. This framework fosters benefits for organizations, society, and the environment, while simultaneously mitigating potential negative consequences. Two perspectives on CDR have been discussed in the literature. Some studies argue that CDR is an extension of Corporate Social Responsibility (CSR) (Herden et al., 2021). Other researchers believe that CDR should be considered as a distinct concept because it explicitly focuses on the creation and use of digital technologies (Elliott et al., 2021; Mihale-Wilson et al., 2022; Roša & Lobanova, 2022). However, it should be noted that similar to CSR, the concept of Corporate Digital Responsibility is also voluntary (Kunz et al., 2024). Organizations have the discretion to integrate it into their culture and approach toward the responsible use of data and digital technologies.

Lobschat et al. (2021) initially defined CDR as "A set of specific values and norms that guide an organization's judgments and choices in matters that specifically relate to digital issues". Herden et al. (2021), on the other hand, argue that considering the ethical opportunities and challenges of digitalization, CDR is an extension of a company's responsibilities. However, based on the available studies, there has been limited research on extracting dimensions and examining indicators for measuring this concept. A significant portion of this article focuses on presenting a measurement model to assess this variable.

On the other hand, the relationship between the adoption of digital technology and sustainable development has not yet been fully elucidated (Jones & Wynn, 2021). For instance, Jones and Wynn (2021) and Awad and Martín-Rojas (2023) argue that digital technologies can effectively assist organizations in achieving sustainability goals. However, other sources indicate that digital technologies have destructive consequences for corporate sustainability, and the risks of unethical digitalization for organizations and their employees are rapidly increasing. An inadequate comprehension of how digitalization impacts sustainability initiatives could constrain businesses' ability to navigate the evolving regulatory landscape and effectively participate in the formulation of policies and regulations governing these technological advancements (Ghobakhloo et al., 2023). George et al. (2021) noted that in addressing critical sustainability challenges, business leaders are increasingly leveraging digital technologies. This approach extends beyond mere technological innovation. It encompasses the development of novel business models that integrate these innovations with a resolute focus on achieving sustainable development goals. He defines digital sustainability as "Organizational activities that seek to advance sustainable development goals through the creative deployment of technologies that create, use, transfer, or store electronic data". Pan and Nishant (2023) assert that Digital enterprises striving towards sustainable development goals prioritize the responsible application of novel digital technologies as a cornerstone of their sustainability commitment. Consequently, adherence to Corporate Digital Responsibility principles, as outlined in its definition, has the potential to reinforce the mutually advantageous convergence of digital transformation and sustainable development, ultimately leading to digital sustainability within organizations (Feroz et al., 2023).

Therefore, the innovative focus of this research is to present a unique measurement model for corporate digital responsibility and identify its main components. In addition, the impact of CDR on the concept of digital sustainability is examined as a significant outcome. Thus, the second section begins with a review of the research background, conceptualizing CDR and its relationship with DS. Subsequently, the third section outlines the methodology employed, while the fourth section details the results obtained from validating the model. In the fifth section, theoretical contributions and managerial applications are discussed.

2. Literature review

2.1. Corporate social responsibility

The most comprehensive corporate responsibility, known as Corporate Social Responsibility (CSR), was first introduced by Bowen (2013). He demonstrated that business managers should not harm the rights and interests of others, emphasizing their responsibility for damages resulting from business (Nguyen et al., 2020). Some scholars argue that social responsibility entails aligning the behavior of a company with societal norms, values, and expectations. Additionally, Carroll (1979) defines corporate social responsibility as "The societal expectations of organizations in terms of economic, legal, ethical, and philanthropic dimensions at a given time". Similarly, Lobschat et al. (2021) describe CSR as "The responsibility of companies to align with the expectations, goals, and values of society and stakeholders". According to CSR, companies must consider the economic, social, and environmental consequences of their actions and, by taking on social responsibility, contribute to improving the quality of life.

Corporate Social Responsibility demands that companies recognize their responsibilities to customers, employees, the environment, and the community, exceeding the pure shareholder focus. Avoiding CSR can lead to a gradual erosion of social power (George et al., 2020). Over the years, CSR has gained significant importance and has recently become a crucial and noteworthy factor for businesses and academics (Manasakis, 2018). Corporate responsibilities and their dimensions guide companies to play their role well as an institution in the society in which they operate to have a share in improving the quality of life of the people of their society and to have a positive impact on their lifestyle.

However, While CSR is widely adopted, its implementation is often criticized due to CSR-washing, where companies project a responsible image without meaningful action (Martin et al., 2024). CSR-washing occurs when companies use sustainability rhetoric for reputational benefits while failing to integrate genuine CSR practices into their core business strategies (Pope & Wæraas, 2016). This discrepancy has fueled skepticism among stakeholders, with some studies suggesting that firms only take CSR seriously when strict regulations and legal consequences are imposed (Crane & Glozer, 2016).

Accountability was historically reinforced by penalties for non-compliance, compelling MNCs to adopt stricter policies (Nietsch, 2018). The enforcement of CSR gained momentum when regulatory bodies imposed substantial fines on corporations found guilty of CSR violations (George et al., 2020). Consequently, CSR became a structured obligation, especially in industries like oil and gas.

In addition, the debate surrounding CSR's effectiveness and its legal versus voluntary nature remains ongoing. Some scholars advocate for stronger legal frameworks, while others support voluntary initiatives. Regardless, studies stress the importance of transparency, stakeholder engagement, and rigorous CSR management (Aguinis & Glavas, 2019).

2.2. Corporate digital responsibility

The evolution of technologies such as artificial intelligence has

fostered the emergence of a novel generation of digital products and services characterized by enhanced intelligence, intricate functionality, and heightened autonomy (Kunz & Wirtz, 2024). They quickly adapt to user needs, seamlessly integrate with other technologies, and deliver more sophisticated products and services (Baird & Maruping, 2021; Mihale-Wilson et al., 2022). While new technologies present new opportunities (Sun et al., 2024), they also bring challenges and new threats to organizations (Švarc et al., 2024). The CSR perspective suggests that, when fulfilling their commitments to stakeholders, companies should minimize harm and maximize their beneficial effects on society (Mihale-Wilson et al., 2022). Recent studies have shown that digital transformation can act as an enabler in improving corporate social responsibility (Song et al., 2025); however, since CSR has a relatively broad societal focus and technology is considered a secondary factor, along with the distinct characteristics of digital technologies compared to other societal dimensions, corporate social responsibility may not adequately respond to the challenges of this specific technological era (Ding et al., 2025).

Digitalization can lead to ethical challenges such as privacy violations of data (Herden et al., 2021) and individuals (Mueller, 2022), and legal challenges, including noncompliance with regional, national, and even corporate laws and regulations (Herden et al., 2021). It can also pose threats against the triple bottom line dimensions (i.e., economic, social, and environmental) (Cardinali & De Giovanni, 2022). In response to these challenges, the concept of "Corporate Digital Responsibility" has been introduced. As one of the most important studies, Lobschat et al. (2021) defined CDR as a set of common values and norms guiding an organization in terms of the four main processes related to digital technology and data, namely technology creation and data collection, operations and decision-making, impact assessment and correction of technology and data; additionally, they argued that CDR should be explicitly considered separate from CSR due to the different characteristics of digital technologies. On the other hand, Herden et al. (2021) considered CDR as an extension of CSR, including all levels of corporate responsibility that are defined in Carroll's CSR pyramid (Carroll, 1991) and all areas of environmental, social, and governance (ESG) framework. Furthermore, Elliott et al. (2021) introduced CDR as a voluntary commitment that informs the role of the company in providing benefits to the community by transparently communicating "good" digital actions and addressing the social, economic, and environmental impacts on the digital society. CDR is proposed as a separate recommendation for organizations specifically engaged with technology and digital data. Independent definitions of CDR have been provided by various studies, such as Mihale-Wilson et al. (2022) and Cheng and Zhang (2023), highlighting the necessity of distinguishing between CDR and CSR.

The growing influence of digital technologies on business operations necessitates the alignment of Corporate Digital Responsibility (CDR) with Sustainable Development Goals (SDGs) and Environmental, Social, and Governance (ESG) frameworks (Herden et al., 2021). As highlighted by Ding et al. (2024), digital transformation can significantly enhance ESG performance through improved green innovation and resource efficiency. CDR strategies should address technological risks while promoting global sustainability, such as minimizing the carbon footprint of digital infrastructures. Integrating CDR with ESG emphasizes transparency, accountability, and ethical governance in areas like AI, privacy protection, and fair data use (Wirtz et al., 2023). CDR also promotes social equity by bridging the digital divide and supports environmental sustainability by addressing issues like the energy consumption of GenAI and blockchain technologies (Azizan et al., 2025; Gopal & Pitts, 2025).

Therefore, not only is there a unified definition for CDR (Herden et al., 2021), but the exact position of CDR in the literature and its relationship with CSR and broader ESG and SDG goals are still subjects of debate among researchers. The lack of consensus on the definition of "corporate digital responsibility" and the necessity of integrating CDR principles with sustainable practices that guide corporate strategies toward responsible and sustainable digital transformation is clearly evident. Therefore, we intend to establish the position of CDR in the literature, considering the emerging challenges and threats and based on previous studies. By providing dimensions and indicators, we intend to present a new definition of this concept, which explicitly incorporates the growing need for alignment with ESG and SDG frameworks.

2.3. CDR elements: dimension identification

As mentioned, this emerging wave of products and services has the potential to not only significantly enhance individual well-being but also unlock new business opportunities. However, it can also pose new ethical, legal, and social issues (Mihale-Wilson et al., 2022). Digital technologies can lead to challenges such as data security threats, data ownership, and privacy issues, increased environmental impact through energy consumption and digital waste (Grybauskas et al., 2022), digital system uncertainties, and discrimination through malicious artificial intelligence (Herden et al., 2021). Based on bibliographic studies, particularly the works of Lobschat et al. (2021) and Herden et al. (2021), we define Corporate Digital Responsibility as "an effort to utilize digital technologies in organizations with an awareness of its impact on itself, society, and the environment in a way that responds to the expectations of stakeholders regarding ethical, legal, and sustainability-oriented issues." We argue that considering that CSR cannot address the challenges and threats of the digital world, CDR is considered a new responsibility that can be seen as an extension of CSR, not as a replacement or an upgraded version but as a new responsibility that parallels CSR and is a subset of corporate responsibility. Previous studies, such as Wirtz et al. (2023), identified ethics, fairness, and privacy as key dimensions of CDR. Our study expands this framework by incorporating legal compliance and sustainability, reflecting the growing importance of regulations and sustainability in shaping corporate digital responsibility. Following this perspective, three dimensions are extracted for CDR.

2.3.1. Ethical digital responsibility

The digitalization of society shifts the boundaries of our capabilities and offers various opportunities, but it also challenges our ethical boundaries (Royakkers et al., 2018). Ethical responsibilities in the digital realm entail avoiding harm and creating benefits for stakeholders through the use of digital technologies. Therefore, digital companies are obligated to pursue ethical actions and conduct themselves excellently, fairly, and justly, surpassing legal frameworks and governance structures to meet the expectations of their stakeholders (Carroll, 1991). Building upon the ethical framework proposed by Wirtz et al. (2023), this study expands their ethical dimension by incorporating additional elements tailored to the challenges of Corporate Digital Responsibility (CDR). While Wirtz et al. focus on fairness, biases, and privacy, our ethical dimension includes data privacy, algorithmic and data biases, and workforce equity, offering a broader perspective.

The first ethical challenge is the protection of individual and data privacy (Mueller, 2022). The ubiquity of technology opens up new ways for businesses to gain profits. However, this profit may come at the expense of privacy. Technologies can track every move, at any time and anywhere, resulting in a massive amount of data being collected about individuals and their activities (Elliott et al., 2021; Mihale-Wilson et al., 2022). Although the potential for data utilization extends to novel business models and cost reductions facilitated by optimized and automated processes, concerns regarding irresponsible data practices concurrently emerge. Responsible data management in terms of data collection and use constitutes the privacy subdimension (Lobschat et al., 2021). Therefore, the concept of CDR voluntarily enhances the protection of privacy and data security beyond regulations and seeks to strengthen consumer trust through a more comprehensive approach than other corporate responsibilities (Wirtz et al., 2023).

The second ethical challenge is the presence of social biases (racial, gender, age, etc.) in data and artificial intelligence-based algorithms

(machine decision-making in general) (Lambrecht & Tucker, 2019; Mueller, 2022). The onus of ethical responsibility extends beyond the sole domain of those who create digital technologies and data, but the data and algorithms themselves can have unethical effects at the operational and decision-making stage (Wirtz et al., 2023). In this regard, Lobschat et al. (2021) state that the outputs and subsequent decisions generated by machine learning algorithms are demonstrably influenced by both the algorithm's structure and the underlying training data. Emerging evidence indicates that utilizing such algorithms, particularly those solely trained on historical data, within legal proceedings or hiring practices can inadvertently perpetuate past racial and gender biases in future outcomes.

To mitigate these biases, the concept of 'Ethics by Design' has gained prominence as a proactive approach to embedding fairness and transparency in algorithmic decision-making (Floridi, 2019). Rather than addressing ethical concerns post-implementation, this principle advocates for the integration of ethical safeguards during the design phase of digital systems. Specifically, Ethics by Design emphasizes the importance of bias-aware algorithm development, diverse and representative training datasets enhance transparency and accountability (Wirtz et al., 2023). Beyond algorithmic structure, data quality plays a crucial role in reducing bias, as incomplete or unrepresentative datasets can reinforce existing inequalities (Mueller, 2022). For instance, credit-scoring models trained on biased financial data may systematically disadvantage underprivileged groups.

The third ethical challenge is discrimination between digital and nondigital employees. Corporate digital responsibility has both an organizational and an interpersonal nature. Employees in a company can interact with each other or outside the organization through digital technologies (Lobschat et al., 2021). In a situation where the main affairs of the organization are carried out through digital technologies (such as enterprise resource planning software), there is a risk of discrimination between employees who use digital technologies and employees who are not required to use these technologies (Floridi, 2010; Lobschat et al., 2021). Therefore, CDR is oriented and serves as a tool to reduce the new form of discrimination between those who need to perform their duties in the digital space and other individuals.

2.3.2. Legal digital responsibility

The inherent characteristics of this emerging generation of systems and services pose a challenge to the established legal boundaries of corporate responsibility (Mihale-Wilson et al., 2022). Thus far, various guidelines and rules have evolved to implement Corporate Digital Responsibility. For instance, formal organizations such as the European Union, OECD, or the United Nations have formulated guidelines and regulations to address various challenges arising from new technologies and digitalization (Baik, 2020). Legal frameworks necessitate compliance with digital technology and data security regulations for companies (Carroll, 1991; Herden et al., 2021).

Organizations are also expected to create and adhere to norms and regulations beyond international, regional, national, and provincial laws and guidelines in the face of dilemmas and challenges related to technology and digital data. Examples include the United States Civil Rights Act of 1964, which addresses employment discrimination (Lambrecht & Tucker, 2019), and the GDPR in the European Union, which provides comprehensive regulations for data protection and privacy (Voigt & Von dem Bussche, 2017). Recent advancements such as the development of artificial intelligence with learning capabilities have further emphasized the need for ethical standards (Kunz & Wirtz, 2024).

Unlike Wirtz et al. (2023), who primarily focus on ethical considerations, our study treats legal compliance as a distinct pillar of CDR. With the increasing role of international regulations and internal corporate governance frameworks, we propose that legal responsibility should be explicitly considered in the evaluation of corporate digital responsibility.

CDR is mainly a voluntary commitment (Elliott et al., 2021), but many scholars argue that voluntary approaches are insufficient to tackle the ethical and societal challenges of digital technologies (Floridi, 2019). For instance, self-regulation alone cannot adequately prevent unintended consequences such as data misuse, algorithmic discrimination, and environmental harm (Mueller, 2022). Floridi (2019) advocates for the introduction of legally binding frameworks that integrate CDR principles into national and international regulations. Regulatory initiatives like the EU's AI Act seek to enforce transparency, accountability, and fairness in AI, shifting from voluntary to enforceable policies (Tosza, 2021). Companies should integrate CDR into internal governance by embedding digital ethics into corporate norms and decision-making to ensure compliance with both external regulations and internal ethical guidelines.

2.3.3. Sustainable digital responsibility

While digital technologies have numerous advantages, they can have unintended negative consequences for the environment and society (Cardinali & De Giovanni, 2022; Chen & Hao, 2022). Emerging technologies increase dynamic competition in the business environment, imposing financial and environmental burdens on manufacturing companies. Unlike Wirtz et al. (2023), our framework integrates sustainability into CDR, aligning corporate digital practices with SDGs and ESG goals.

The environmental impact of digital technologies, such as Generative AI and blockchain-based cryptocurrency mining, raises significant concerns regarding energy consumption and sustainability (Gopal & Pitts, 2025). Recent reports have highlighted that the computational power required for training AI models and mining cryptocurrencies results in substantial energy use, contributing to carbon emissions and environmental degradation. These unfavorable environmental conditions may lead to an increased demand for energy and resources, ultimately resulting in greater pollution and waste generation (Ghobakhloo, 2020; Kruk et al., 2023). Therefore, organizations need to act responsibly and conscientiously regarding the detrimental effects of new digital technologies on waste (Ullah et al., 2024). Organizations must mitigate these effects by reducing energy use, adopting clean energy, and optimizing digital technologies under CDR principles (Mihale-Wilson et al., 2022).

In the current era, there is a concern for organizations that have maintained their traditional processes, as they may lose their competitive advantage and innovation over time due to the competitive pressures of the present age and may not be able to meet the expectations and economic needs of their stakeholders (Hendriarto, 2021). The digitalization of organizations can lead to the discovery of new business opportunities and also enable new value propositions (Ritter & Pedersen, 2020). CDR should guide organizations on how to manage potential negative consequences and leverage digital opportunities (Lobschat et al., 2021). Therefore, organizations bear the responsibility of creating technology-based innovations in the face of competitive pressure (Memon & Ooi, 2023).

Concerns about sustainability are perhaps the only points of overlap between CSR and CDR (Mihale-Wilson et al., 2022). According to CSR principles, companies should consider the economic, social, and environmental impacts of their actions (Aguinis, 2011). As shown empirically in Cardinali and De Giovanni (2022), digital technologies can damage both the social and environmental domains and threaten the sustainability of the organization and its environment. Therefore, CDR obliges organizations to use and share data and digital technologies to achieve sustainable development goals.

In addition, the rise of AI and automation threatens job security, with employees fearing displacement due to a lack of digital skills (Chen et al., 2022). Jobs are disappearing both in industry and in the service sector. Many employees in organizations that are moving to digitalization are afraid of losing their jobs due to a lack of digital skills (Herden et al., 2021). As jobs disappear across industries, organizations must proactively create new roles and upskill employees to adapt to technological advancements (Grybauskas et al., 2022). Therefore, The

challenge lies not in resisting digitalization but in strategically aligning workforce development with technological progress. CDR emphasizes employee readiness, urging organizations to balance digital transformation with skill-building and ensure a sustainable transition (Elliott & Copilah-Ali, 2024). Moreover, it is essential to consider and cultivate the potential of employees in the adoption and development of new digital technologies.

2.4. CDR and digital sustainability

According to the dimensions and the definition provided, we intend to examine the relationship between CDR and digital sustainability. Digital Sustainability and Corporate Digital Responsibility are often addressed separately in the literature, missing potential synergies (Covucci et al., 2024). Currently, due to the digitalization of businesses, managers are trying to address the critical challenges of sustainability by using digital technologies (George et al., 2021; Lin & Zhang, 2023). Sustainability encompasses three pillars: environmental, social, and economic (Wut et al., 2021). Environmental sustainability is defined as the continuous preservation of the ecosystem and its functioning. Economic sustainability refers to the system's ability to sustainably meet all economic expectations. Social sustainability pursues the well-being of all stakeholders. Digitalization leads to various challenges in sustainability, such as increased electricity consumption, cybersecurity threats (Al-Khateeb et al., 2017), and social conflicts (Linkov et al., 2018). For a digital ecosystem to be sustainable, its governance needs to be adaptable and responsive to the cultural background, traditional practices, and specific needs of the organizations involved (Ntim & Soobaroyen, 2013). Therefore, Corporate Digital Responsibility, as an upstream responsibility in the organization and considering the provided definition and its dimensions, can guide digital organizations toward Digital Sustainability.

As digital technologies impact business operations, Corporate Digital Responsibility (CDR) helps organizations integrate responsible practices while achieving sustainability goals (Herden et al., 2021). CDR aligns digital initiatives with environmental, social, and governance (ESG) principles, guiding ethical digital practices and ensuring long-term sustainability without negative impacts. It balances innovation with sustainable development. Additionally, with the aim of this research and after extracting the dimensions and indicators of CDR, a conceptual model (Fig. 1) was obtained, which will be further analyzed in the following section.

3. Methodology

This study aims to establish dimensions and indicators for measuring Corporate Digital Responsibility (CDR) in organizations and assess its impact on digital sustainability. To achieve this, a multi-stage methodology was employed. First, dimensions and indicators were identified through an extensive literature review and further refined using the Delphi method. In the subsequent stages, exploratory factor analyses (EFA) and confirmatory factor analyses (CFA) were employed to ensure construct validity and test the proposed model.

SPSS 26 was used to perform **Exploratory Factor Analysis**, which helped evaluate and confirm the three dimensions of CDR. The Kaiser-Meyer-Olkin (KMO) test yielded a value of 0.898, confirming sampling adequacy and the suitability of the data for factor analysis (Hair et al., 2013). EFA was conducted using principal component analysis with varimax rotation to identify the latent structure of the measurement model. The results indicated that the 11 indicators grouped logically into three dimensions, consistent with the theoretical foundations derived from the literature and refined through the Delphi method.

In the final stage, **Smart-PLS 3.0** was utilized to perform **Confirmatory Factor Analysis** and validate the proposed research model. The structural equation modeling (SEM) approach based on variance was chosen due to its alignment with the objectives of this study (Hair et al., 2012). CFA was employed to confirm the dimensional structure of the CDR construct and to evaluate the reliability and validity of the measurement model. This analytical approach ensured that both convergent and discriminant validity were systematically assessed, providing robust empirical support for the framework. The empirical methodology employed in this study enhances the theoretical contributions to CDR and addresses a significant gap in the literature by offering a validated measurement instrument.

3.1. Measurements

3.1.1. Measure of CDR

In order to discover the dimensions and indicators for measuring the concept of CDR we used research in the field of corporate responsibility, especially corporate social responsibility and corporate digital responsibility; the challenges of digital transformation; new generations of industry; and corporate ethics. Therefore, after studying the necessary resources and based on the literature, 11 items were identified in 3 sections to measure CDR.

To confirm and finalize the indicators, a two-round Delphi method was employed, which is a widely accepted approach for achieving expert

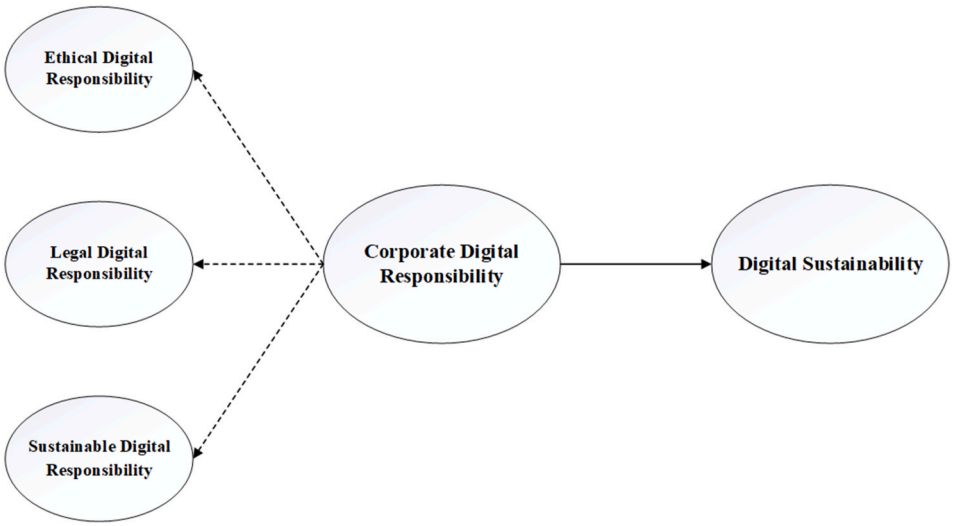


Fig. 1. Conceptual model.

consensus in exploratory and novel studies (Hsu & Sandford, 2007; Okoli & Pawlowski, 2004). A structured questionnaire was developed, including two parts: the first part comprised the 11 indicators across 3 dimensions rated on a five-point Likert scale, and the second part included two open-ended questions (Tran et al., 2020). Purposive sampling was used in this study to intentionally select individuals with specific characteristics or experiences relevant to the research (Memon et al., 2024). This approach ensures the collection of rich, in-depth data, enhancing the quality and relevance of the findings (Campbell et al., 2020). Subsequently, 29 people whose research background was relevant to the field of this research according to the ISI Web of Science/Scopus or whose specialized field of work aligned with the scope of this research were selected and the questionnaire was sent to them. Finally, 23 people agreed to participate in the first round of research (acceptance rate 79 %). According to Hsu and Sandford (2007), this number of participants was satisfactory.

The use of the Delphi method ensured content validity by involving domain experts in iterative rounds of review. Participants were asked to evaluate each item for clarity, relevance, and representativeness of the proposed dimension. Items that did not reach a minimum agreement threshold were revised or removed, following the guidance of Keeney et al. (2006). According to the criteria of Keeney et al. (2006), the second round of questionnaire distribution was conducted. The second round of the questionnaire included the same 13 initial questions along with suggested changes (including changes in the name of the indicator or its definition) from experts in the first round (see Appendix A). Therefore, the questionnaire was sent to the same 23 people, and only 20 people participated. After the data were collected, all 11 indicators reached a consensus, and no additional items were added. Therefore, the dimensions and measurement indicators were finalized after two rounds of the Delphi study with some changes. The results are shown in Appendix B.

The scales used in this study are self-constructed and developed specifically for the purposes of this research. While the indicators and dimensions were designed independently, the underlying framework was informed by existing literature on Corporate Digital Responsibility. The relevant theories and prior studies provided a foundation for the identification of key dimensions, but the actual scales were created by the researchers to fit the specific context and objectives of this study.

To ensure the robustness of the developed instrument, multiple statistical techniques were applied. Content validity was established via a two-round Delphi method with expert feedback. Construct validity was assessed through Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA). The KMO value confirmed sampling adequacy, and factor loadings were evaluated for indicator reliability. Furthermore, convergent validity was tested using Average Variance Extracted (AVE), and internal consistency was assessed via Cronbach's Alpha and Composite Reliability (CR). These validation procedures, detailed in the Results section, provided empirical support for the reliability and validity of the newly developed measurement model.

3.1.2. Measure of digital sustainability

To measure digital sustainability, we used the studies of Wut et al. (2021) and George et al. (2021). In total, 8 questions were obtained to assess this variable (Appendix C) and were incorporated into the questionnaire. DS was measured on a five-point Likert scale ranging from 1 "Strongly Disagree" to 5 "Strongly Agree". This scale gauges the perceptions of employees and managers in digital businesses regarding the digital sustainability of their companies.

3.2. Sample and data

According to the research topic, the target population in this study needed to be selected so that they possessed the competence to respond to the raised questions. Therefore, it was necessary to select companies with a high level of technology since they have more experience in digitalization and the use of new digital technologies. It was decided to collect data from countries whose digital competitiveness score in the 2022 digital competitiveness ranking was above 85. This selection would homogenize the target population selected and make the results reliable. Therefore, a list of companies that used digital technologies in their processes or products was prepared. In this study Random sampling was employed to ensure a representative selection of participants, minimizing bias and allowing for the generalization of findings to the broader population (Creswell & Creswell, 2017). This method enhances the external validity of the study by providing a fair and unbiased representation of the target group (Fowler Jr, 2013).

Initially, the questionnaire was distributed to employees at 15 companies to assess their understanding of the survey questions. Twelve responses were received, and the results of this pilot study were utilized to improve and revise the questions. Next, 410 companies were randomly selected, and emails containing an introduction and the questionnaire were sent to eligible employees or managers. Data were collected over a limited three-month period. Furthermore, the target population was not easily accessible, which made the process of collecting questionnaires particularly challenging. Throughout the data collection process, ethical considerations were prioritized. All Participants were informed of the study's purpose, assured of voluntary participation, and provided informed consent. To protect confidentiality, personal information remained anonymous, and data were securely stored. These measures ensured participant privacy and data security. Eventually, 145 completed questionnaires were collected, resulting in a response rate of 35.3 %. The demographic statistics indicated a relative balance, with 60.0 % of the respondents being male and 40.0 % being female, contributing to the prevention of potential bias. In terms of education, 36.6 % held a bachelor's degree, 34.5 % held a master's degree, and 6.2 % had completed a doctoral degree. Thus, 77.3 % of the respondents had a bachelor's degree or higher, indicating a sufficient level of expertise to respond to the questionnaire's inquiries (Table 1). The statistical distribution also highlighted the diverse experiences of the respondents, ranging from small to large organizations.

Table 1
Demographic statistics of the respondents.

Category	Description	Frequency (145)	Percentage %
Gender	Male	87	60.0
	Female	58	40.0
Degree of education	Diploma & Under Diploma	24	16.6
	Associate	9	6.2
	Bachelor	53	36.6
	Master	50	34.5
	Ph.D	9	6.2
Job level	Employee or Expert	94	64.8
	Middle Manager	31	21.4
	Top Manager	20	13.8
Company size	1 to 9 persons	25	17.2
	10 to 49 persons	29	20.0
	50 to 249 persons	28	19.3
	250 to 500 persons	19	13.1
	More than 500 persons	44	30.3
Field of company	Financial and trade services	18	12.4
	High Tech	38	26.2
	Retail and customer service	28	19.3
	Automotive and household appliances	28	19.3
	Oil, gas, and minerals	12	8.2
	Food and Agriculture	11	7.6
Company location	Europe	88	60.7
	North America	26	17.9
	Asia	24	16.6
	Australia	7	4.8

4. Results

This section presents the findings of the study, focusing on the validation and assessment of the proposed Corporate Digital Responsibility (CDR) framework. The results are organized into three main parts: (1) exploratory factor analysis (EFA) to confirm the dimensions of CDR, (2) measurement model evaluation to assess reliability and validity, and (3) conceptual model assessment to evaluate the impact of CDR on digital sustainability. These analyses provide empirical support for the robustness of the CDR framework and its significant role in promoting digital sustainability within organizations.

4.1. Exploratory factor analysis

The exploratory factor analysis results confirmed the existence of three distinct dimensions for Corporate Digital Responsibility, with all 11 indicators grouped into these categories. This alignment between the EFA outcomes and the dimensions identified in the literature review and Delphi method supports the robustness of the proposed model. The three dimensions provide a comprehensive framework for measuring CDR in organizations, each reflecting critical aspects of ethical, legal, and sustainability-oriented responsibilities.

As shown in Table 2, the categorization of the indicators aligns with the theoretical foundations of CDR, ensuring consistency between empirical findings and the conceptual model. This validation highlights the applicability of the proposed dimensions across organizational contexts and sets a foundation for further research into their impact on digital sustainability.

4.2. Measurement model

The values of the mean and standard deviation for each of the indicators are shown in Table 3. The measurement model was evaluated by examining three criteria: factor loadings, reliability, and validity. The first criterion for assurance involved inspecting the factor loadings of individual items for each dimension. All factor loadings of the indicators were significantly greater than the proposed acceptance level of 0.7 (Borah et al., 2022). Cronbach's alpha and composite reliability were also calculated, with an acceptance level of 0.70 used to assess reliability (Hair et al., 2019). Cronbach's alpha and composite reliability for all the constructs were found to be above 0.70, indicating satisfactory reliability in this study. Furthermore, Cronbach's alpha and composite reliability cannot exceed 0.95, because in this case, these indicators measure the same concept (Wut et al., 2021). In this research, convergent validity was assessed using the average variance extracted (AVE), with a threshold of 0.50. The AVE values for all three dimensions were greater than 0.50, confirming convergent validity (Fornell & Larcker, 1981; Hair et al., 2019).

To meet the requirements of discriminant validity, the square root of the AVE for a construct should be greater than the correlation between that construct and the other constructs in the model (Chen, 2011).

Table 2
Exploratory factor analysis of CDR.

Item	Ethical Digital Responsibility	Legal Digital Responsibility	Sustainable Digital Responsibility
Q1	**0.795**	0.165	0.330
Q2	**0.843**	0.189	0.206
Q3	**0.842**	0.202	0.282
Q4	0.354	**0.748**	0.236
Q5	0.128	**0.841**	0.294
Q6	0.316	0.225	**0.724**
Q7	0.257	0.316	**0.721**
Q8	0.203	0.154	**0.743**
Q9	0.166	0.125	**0.833**
Q10	0.330	0.189	**0.670**
Q11	0.179	0.219	**0.742**

Table 3
Reliability and average variance extracted for construct.

Construct/ Indicators	Mean	SD	Factor Loading	Cronbach's Alpha	CR	AVE
Ethical Digital Responsibility				0.873	0.922	0.798
Q1	4.14	0.790	0.891			
Q2	4.19	0.866	0.875			
Q3	4.18	0.847	0.913			
Legal Digital Responsibility				0.755	0.891	0.803
Q4	4.14	0.855	0.908			
Q5	3.75	0.829	0.884			
Sustainable Digital Responsibility				0.891	0.917	0.648
Q6	3.81	0.908	0.823			
Q7	3.78	0.982	0.836			
Q8	3.58	0.918	0.775			
Q9	3.60	0.996	0.832			
Q10	3.86	0.913	0.778			
Q11	3.74	0.970	0.784			
Digital Sustainability				0.890	0.912	0.564
Q12	4.18	0.887	0.655			
Q13	4.35	0.741	0.704			
Q14	4.38	0.646	0.779			
Q15	4.16	0.831	0.784			
Q16	4.01	0.954	0.777			
Q17	3.92	0.917	0.794			
Q18	3.79	0.899	0.754			
Q19	3.81	0.913	0.753			

According to the results presented in Table 4, the principal diagonal values of the matrix; or the square root of the AVE are greater than the correlation coefficients of the other constructs in the model; therefore, the discriminant validity of the model is appropriate according to the Fornell-Larcker criterion. Consequently, the measurements in this study demonstrated satisfactory convergent and discriminant validity. Based on several reliability and validity tests, this study has sufficient reliability and validity.

4.3. Conceptual model assessment

With the emerging concept of CDR and the various hypotheses that have been expressed about its impact on organizations, the structural model of this study shows the impact of this concept on DS. To evaluate this model and prove the above hypothesis, path coefficients, significance levels, coefficients of determination (R^2), and predictive power indices (Q2) were used as indicators of the fit of the structural equation model (Fig. 2). The significance level was assessed using the bootstrapping test in PLS software with 500 subsamples, calculating the t-value. In this study, the proposed hypothesis was accepted with a t-value higher than 1.96 and a p-value less than 0.05 (t-value = 17.427).

The R^2 values are considered weak, medium, and significant with values of 0.19, 0.33, and 0.60, respectively (Cohen, 1992). In this study, R^2 was equal to 0.482, which indicates that the impact of CDR on DS is significant. In addition, since SRMR = 0.111, structural model validity is confirmed, which also enhances the validity of the measurement model for CDR. Values of Q^2 above zero indicate that the values obtained by the model are well-reconstructed and that the model has predictive capability (Chin & Marcolides, 1988). Similarly, the predictive power is weak, medium, and large at thresholds of 0.02, 0.15, and 0.35, respectively. The current study's model demonstrates a significant predictive capability for digital sustainability, with a value greater than 0.247. In this study, Harman's single-factor test was conducted to assess the presence of common method bias (CMB). The statistical analysis indicates that the first factor explains only 46.16 % of the total variance of the variables. This suggests that CMB did not significantly influence the findings (MacKenzie & Podsakoff, 2012).

Table 4
Fornell–Larcker criterion.

	Ethical Digital Responsibility	Legal Digital Responsibility	Sustainable Digital Responsibility	Digital Sustainability
Ethical Digital Responsibility	0.893			
Legal Digital Responsibility	0.525	0.896		
Sustainable Digital Responsibility	0.610	0.554	0.805	
Digital Sustainability	0.537	0.555	0.649	0.751

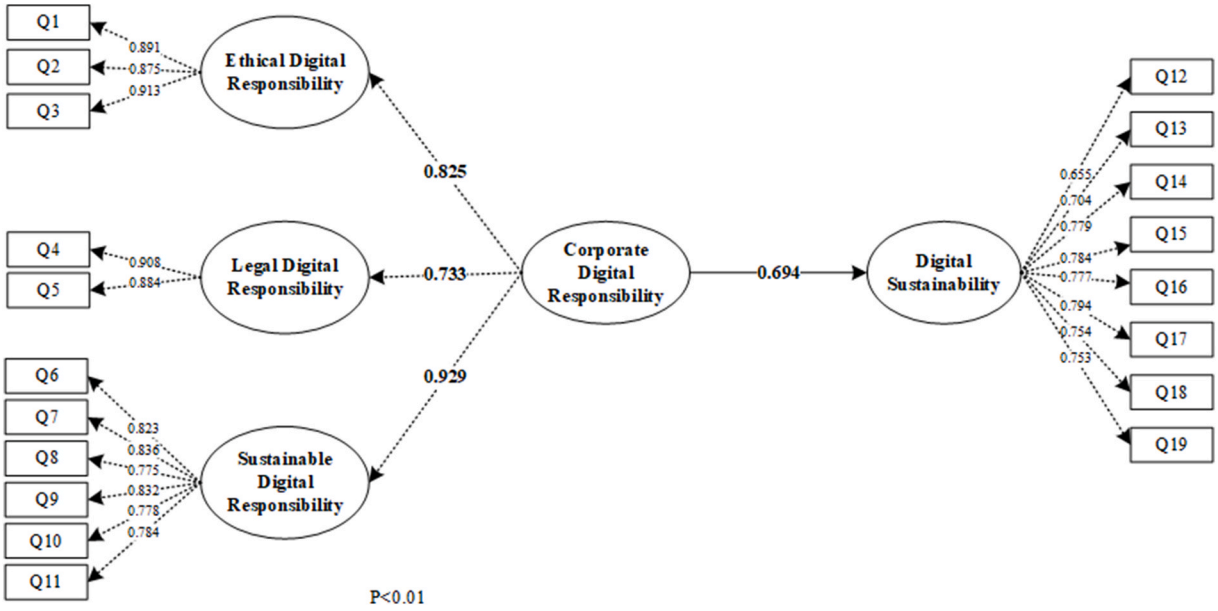


Fig. 2. Results.

5. Discussion and conclusion

A review of the literature showed that the challenges of digitalization have led to the emergence of CDR as the newest corporate responsibility. It was also evident that there are limited scales available to measure this concept in organizations (Carl et al., 2023; Cheng & Zhang, 2023). In this study, for the first time, we defined and expanded the CDR concept in the form of three dimensions: ethical digital responsibility, legal digital responsibility, and sustainable digital responsibility. The findings of this study, while novel, align with previous research on digital responsibility and digital sustainability (e.g., George et al. (2021)). However, this study further contributes to the literature by introducing a structured framework for Corporate Digital Responsibility, validated through both exploratory and confirmatory analyses. This dual validation ensures the robustness of the proposed dimensions, highlighting their practical relevance.

The Delphi method facilitated expert consensus to refine CDR dimensions, though the diversity of participants may limit generalizability of the findings. While exploratory and confirmatory factor analyses validated the framework, reliance on self-reported data introduces potential biases. Despite these limitations, the findings contribute significantly to the literature by offering a structured approach to integrating ethical, legal, and sustainability dimensions into digital practices.

5.1. Theoretical contributions

In this study, we described CDR from the perspective of its position relative to CSR and the new challenges arising from the digitalization of products and processes. Previous studies have also highlighted the importance of elements such as ethical dilemmas, including data privacy (Mueller, 2022); social biases (Lambrecht & Tucker, 2019; Lobschat et al., 2021); discrimination and unequal access (Floridi, 2010); legal dilemmas, such as regional standard laws or local and organizational laws for data protection (Herden et al., 2021; Voigt & Von dem Bussche, 2017); and challenges in the area of economic, social, and environmental sustainability (Chen et al., 2022; Hendriarto, 2021; Mihale-Wilson et al., 2022), in CDR. This study shows that, in organizations, CDR focuses on utilizing digital technologies to meet stakeholder expectations, while also addressing the challenges and dilemmas arising from this fundamental change by adopting an ethical, legal, and sustainable approach to digital business development. Furthermore, based on the results, digital companies can leverage their digital advantages and enhance digital sustainability by adhering to CDR principles to achieve their goals. In this regard, Weber-Lewerenz (2021) states that compliance with CDR when using digital technologies and artificial intelligence and launching a secure digital transformation environment for secure management of corporate data and protection against abuse will lead to sustainability. This issue has also been mentioned in other studies such as Trittin-Ulbrich and Böckel (2022), in terms of the relationship between CDR and DS and the positive impact between the two concepts.

These findings will make a significant contribution to the theoretical research on CDR and will fill the existing research gaps for researchers. First, as mentioned, the concept of corporate digital responsibility was developed by proposing a three-dimensional model with indicators for each of the dimensions. Second, based on the literature on corporate responsibilities, the position of this concept relative to other corporate responsibilities was shown, and this work will expand future research in the field of corporate responsibility. Third, the impact of adhering to CDR in an organization on moving toward digital sustainability was investigated, and the results indicate that these two variables are positively related.

5.2. Managerial implications

This study not only helps experts but also has implications for business owners and senior managers, as it emphasizes the crucial goal of companies to act responsibly in society. Achieving this goal requires changes in policies, frameworks, and organizational culture, relying on the support and beliefs of senior managers (Chen & Hao, 2022). This study introduces corporate digital responsibility as a new form of corporate responsibility in the digital age, highlighting key dilemmas in three categories. This research reveals the significant impact of CDR on digital sustainability (DS) in organizations, emphasizing the need for serious consideration by business managers in the responsible and purposeful use of digital technologies for sustainable success. Policymakers are advised to prioritize ethical digital responsibility to meet stakeholder expectations, positively influencing perspectives and enhancing digital sustainability within organizations.

Based on the findings of this research, it is recommended that managers and leaders of organizations consider the following recommendations in their decision-making and policy setting. First, when digital technologies are used, privacy protection should be achieved by establishing appropriate rules and processes, particularly when individuals are involved. Second, not to neglect the training of their human resources in the face of the rapid expansion of the use of digital tools and the digitalization of the organization, and as an important part of their capital and one of their key stakeholders, develop their digital skills in the direction of their talents. Additionally, they do not consider the use or non-use of digital technologies as a basis for comparison or promotion by employees and do not make them the basis for performance assessment. Third, to prevent ethical issues related to bias in machine and AI-based decision-making, in addition to following existing laws at various levels and creating intra-organizational rules and norms, it is necessary to act carefully in the design and development of algorithms and use teams from different backgrounds. Also, use unbiased data (data that is large enough, comprehensive, and free of discriminatory patterns in terms of gender, race, age, etc.) to train the algorithms. The proposed CDR framework highlights data privacy as a key aspect of ethical responsibility; however, achieving completely unbiased data remains a challenge due to inherent biases in data creation and system design. While transparency, inclusivity, and fairness can help mitigate bias, they cannot fully eliminate it. Organizations should adopt measures such as algorithm audits and diverse datasets to address these issues.

Additionally, digital technologies should be leveraged to enhance business models and maintain competitive advantage. The adoption of clean and green technologies can help minimize industrial pollution by integrating renewable energy sources. To reduce the environmental impact of digital infrastructure, companies should invest in energy-efficient algorithms, sustainable AI and blockchain solutions, carbon offset initiatives, and decentralized models with lower computational demands.

5.3. Limitations and future research directions

This study acknowledges several limitations. First, due to the rapid evolution of technology, new challenges related to digital technologies may emerge, necessitating continuous adaptation. While the findings align with previous research, they extend the literature by presenting a structured CDR framework. However, the shortage of professionals trained in 'ethics by design' remains a challenge in AI decision-making, necessitating the integration of ethical AI training into corporate policies. Second, focusing research on a single country or industry could improve the study's credibility. Third, the research focused on one aspect of CDR's impact on digital sustainability, and future studies should explore other outcomes and concepts that foster organizational acceptance of CDR. Fourth, purposive sampling and the Delphi method's reliance on expert diversity limit generalizability, and self-reported data may introduce biases. Future studies should expand the sample size and explore longitudinal approaches. Fifth, while this study addressed privacy, algorithmic biases, and workforce equity, other critical issues like dehumanization and digital addiction should be examined further. Lastly, future research could explore the financial challenges associated with CDR, particularly regarding energy consumption in AI and blockchain systems. Additionally, investigating a CDR maturity model could provide a structured framework for assessing how organizations progress in implementing CDR principles and identifying key stages for improvement.

CRediT authorship contribution statement

Ali Khani: Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Mohsen Shahriari:** Writing – review & editing, Validation, Project administration, Methodology, Conceptualization. **Gadaf Rexhepi:** Writing – review & editing, Validation, Supervision, Methodology.

Declarations of competing interest

None.

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Appendix A

Table 5
Delphi Full Questionnaire

No.	Questions (second round)	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
1	**Privacy and security** First-round Average responses = 4.67 While adopting and developing technology, data collection, and using data, Digital Organizations are facing challenges in protecting personal data and preserving privacy.					
2	**Social biases in decision-making** First-round Average responses = 4.47 By increasing the use of technology, digital leaders are exposed to biases and errors, such as racial and gender biases, in machine learning and AI-based decision-making (biases could be introduced either by their creators or by the dataset used to train the systems themselves).					

(*continued on next page*)

Table 5 (*continued*)

No.	Questions (second round)	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
3	**Discrimination among employees** First-round Average responses = 4.13 The creation and exploitation of digital technology and data can create new forms of discrimination between those required to use digital technologies and other employees. In this way, these people in the organization will have more respect, benefits, and authority just because of the use of digital technologies.					
4	**Law and norm (Compliance with existing laws)** First-round Average responses = 4.53 The use of technologies and digital data has created challenges that require organizations to face ethical issues related to technology and the protection of digital data, requiring existing national and regional legal frameworks.					
5	**Law and norm (Beyond existing laws)** First-round Average responses = 4.53 The use of digital technologies and data has caused challenges that have made organizations require a set of rules even beyond legal frameworks, in the face of ethical issues related to technology and digital data protection.					
6	**Human resources challenge (Employees admission)** First-round Average responses = 4.27 In the process of digital transformation, jobs are changing in a way that requires new skills and talent development. Therefore, digital leaders are faced with the challenge of losing talented employees due to a lack of necessary digital skills and talent.					
7	**The challenge of human resources (talents)** First-round Average responses = 4.20 The rapid pace of digitalization creates a fear of job loss and a lack of necessary preparation among employees, which can lead to resistance and a lack of support for digital transformation.					
8	**Digital technology waste** First-round Average responses = 4.47 Digital waste or digital footprint includes electronic waste, non-recyclable waste, carbon footprint, greenhouse gases, etc contributing to climate change and damage to the natural environment.					
9	**Natural resources and energy** First-round Average responses = 4.53 Many digital technologies require significant energy (i.e. electrical energy) to process and execute tasks. Additionally, some infrastructures used in new digital technologies have increased the consumption of natural resources and materials.					
10	**Economic challenges** First-round Average responses = 4.33 In today's world and due to the new competitive pressures in the digital world, traditional business models cannot create or maintain a sustainable competitive advantage, and the organization					
11	**Sustainable development** First-round Average responses = 4.33 Considering that the goal of many companies is sustainable development, moving towards the use of digital technologies, sharing knowledge, and using data and new technologies in ways that make sustainable development possible are among the challenges that organizations face.					
12	**If there are any further issues or concerns please let us know. You can add them here: (In case needed, please contact us for more details. E-mail:** Business.Lab.digital@gmail.com**)**					
13	**If there are any questions needed to be modified or adjusted, please note them below.**					

Appendix B

Table 6
CDR Measures and Items

Construct		Indicators
Ethical Digital Responsibility		– Our company considers data security and people's privacy in adopting and developing technology, data collection, management, and use of data. – Our company is careful not to have racial, gender, etc. biases in artificial intelligence-based and automated decision-making. – Our company does not discriminate between employees who need to use digital technologies and other employees
Legal Digital Responsibility		– Our company adheres to established legal frameworks governing digital technology and data security. – Our organization has established a robust and cohesive framework of ethical principles to navigate dilemmas associated with technology and digital data.
Sustainable Digital Responsibility	Social	– our company pays enough attention to employees' readiness and acceptance in making decisions on the pace of organizational digitalization and adopting new digital technologies. – our company always considers employees' talent in applying and developing new digital technologies and nurtures them in the right direction.
	Environmental	– our company acts knowingly and responsibly against the destructive effects of new digital technology waste. – In developing new digital technologies, our company has a responsible approach to reducing energy consumption and using green energy.
	economic	– While facing competitive pressures, our company emphasizes creating innovations based on digital technologies. – Our company is committed to using and sharing data and new technologies for sustainable development.

Appendix C

Table 7
Digital Sustainability Items

Construct	Indicators
Digital Sustainability (George et al., 2021; Wut et al., 2021)	– In our company, documents are converted into digital Content (photos, files, data, etc.) with the ability to be used in the long term. – Our company uses technology to store and document digital content. – Digital technologies are maintained and supported in our company. – Our company provides the security of employees' digital resources in order to support digital content. – Our company provides technology for employees to update, share and publicly access digital content. – Our company carries out activities that lead to the development and spread of interaction with digital content. – Our company has an environmentally friendly approach to the development of digital activities. – Our company considers the interests of employees and society in developing its content and digital activities.

Availability of data

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

References

Aguinis, H. (2011). *Organizational responsibility: Doing good and doing well*.

Aguinis, H., & Glavas, A. (2019). On corporate social responsibility, sensemaking, and the search for meaningfulness through work. *Journal of Management, 45*(3), 1057–1086.

Al-Khateeb, H. M., Epiphaniou, G., Alhaboby, Z. A., Barnes, J., & Short, E. (2017). Cyberstalking: Investigating formal intervention and the role of corporate social responsibility. *Telematics and Informatics, 34*(4), 339–349.

Awad, J., & Martín-Rojas, R. (2023). Enhancing social responsibility and resilience through entrepreneurship and digital environment. *Corporate Social Responsibility and Environmental Management*.

Azizan, O., Tajmir Riahi, M., Shahriari, M., & Rasti-Barzoki, M. (2025). The effect of green culture and identity on organizational commitment. *Journal of Environmental Planning and Management, 68*(4), 843–865.

Baik, J. S. (2020). Data privacy against innovation or against discrimination?: The case of the California Consumer Privacy Act (CCPA). *Telematics and Informatics, 52*.

Bailey, A., & Ngwenyama, O. (2011). The challenge of e-participation in the digital city: Exploring generational influences among community telecentre users. *Telematics and Informatics, 28*(3), 204–214.

Baird, A., & Maruping, L. M. (2021). The next generation of research on IS use: A theoretical framework of delegation to and from agentic IS artifacts. *MIS Quarterly, 45*(1).

Basu, S., Jha, S., Ooi, K.-B., & Cham, T.-H. (2025). Unveiling the potential of corporate digital responsibility (CDR). *Journal of Computer Information Systems*, 1–33.

Borah, P. S., Iqbal, S., & Akhtar, S. (2022). Linking social media usage and SME's sustainable performance: The role of digital leadership and innovation capabilities. *Technology in Society, 68*, Article 101900.

Bowen, H. R. (2013). *Social responsibilities of the businessman*. University of Iowa Press.

Campbell, S., Greenwood, M., Prior, S., Shearer, T., Walkem, K., Young, S., Bywaters, D., & Walker, K. (2020). Purposive sampling: Complex or simple? Research case examples. *Journal of Research in Nursing, 25*(8), 652–661.

Cardinali, P. G., & De Giovanni, P. (2022). Responsible digitalization through digital technologies and green practices. *Corporate Social Responsibility and Environmental Management, 29*(4), 984–995.

Carl, K. V., Mihale-Wilson, C., Zibuschka, J., & Hinz, O. (2023). A consumer perspective on Corporate Digital Responsibility: An empirical evaluation of consumer preferences. *Journal of Business Economics*, 1–46.

Carroll, A. B. (1991). The pyramid of corporate social responsibility: Toward the moral management of organizational stakeholders. *Business Horizons, 34*(4), 39–48.

Chen, Y. S. (2011). Green organizational identity: Sources and consequence. *Management Decision, 49*(3), 384–404.

Chen, P., & Hao, Y. (2022). Digital transformation and corporate environmental performance: The moderating role of board characteristics. *Corporate Social Responsibility and Environmental Management, 29*(5), 1757–1767.

Chen, N., Li, Z., & Tang, B. (2022). Can digital skill protect against job displacement risk caused by artificial intelligence? Empirical evidence from 701 detailed occupations. *PLoS One, 17*(11), Article e0277280.

Cheng, C., & Zhang, M. (2023). Conceptualizing corporate digital responsibility: A digital technology development perspective. *Sustainability, 15*(3), 2319.

Chin, W., & Marcolides, G. (1988). Partial least square to structural equation modelling. *Modern Methods for Business Research*. California: LA Wrence Erlbaum Associate.

Cohen, J. (1992). Statistical power analysis. *Current Directions in Psychological Science, 1*(3), 98–101. https://doi.org/10.1111/1467-8721.ep10768783

Covucci, C., Confetto, M. G., Kljucnikov, A., & Panait, M. (2024). Unrevealing the nexus between digital sustainability and corporate digital responsibility: A dual-track systematic literature review towards a framework of corporate digital sustainability. *Technology in Society*, Article 102743.

Crane, A., & Glozer, S. (2016). Researching corporate social responsibility communication: Themes, opportunities and challenges. *Journal of Management Studies, 53*(7), 1223–1252.

Creswell, J. W., & Creswell, J. D. (2017). *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications.

Ding, X., Appolloni, A., Shahzad, M., Liu, Y., & Han, S. (2025). Digital transformation and total factor productivity in manufacturing firms: Evidence of corporate public responsibilities in China. *Technology in Society, 82*, Article 102874.

Ding, X., Vuković, D. B., Sokolov, B. I., Vukovic, N., & Liu, Y. (2024). Enhancing ESG performance through digital transformation: Insights from China's manufacturing sector. *Technology in Society, 79*, Article 102753.

Elliott, K., & Copilah-Ali, J. (2024). Implementing corporate digital responsibility (CDR): Tackling wicked problems for the digital era: Pilot study insights. *Organizational Dynamics*, Article 101040.

Elliott, K., Price, R., Shaw, P., Spiliotopoulos, T., Ng, M., Coopamootoo, K., & van Moorsel, A. (2021). Towards an equitable digital society: Artificial intelligence (AI) and corporate digital responsibility (CDR). *Society, 58*(3), 179–188.

Feroz, A. K., Zo, H., Eom, J., & Chiravuri, A. (2023). Identifying organizations' dynamic capabilities for sustainable digital transformation: A mixed methods study. *Technology in Society, 73*, Article 102257.

Floridi, L. (2010). Ethics after the information revolution. In *The Cambridge handbook of information and computer ethics* (pp. 3–19).

Floridi, L. (2019). Translating principles into practices of digital ethics: Five risks of being unethical. *Philosophy & Technology, 32*(2), 185–193.

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research, 18*(1), 39–50.

Fowler Jr, F. J. (2013). *Survey research methods*. Sage publications.

George, N. A., Aboobaker, N., & Edward, M. (2020). Corporate social responsibility and organizational commitment: Effects of CSR attitude, organizational trust and identification. *Society and Business Review, 15*(3), 255–272.

George, G., Merrill, R. K., & Schillebeeckx, S. J. (2021). Digital sustainability and entrepreneurship: How digital innovations are helping tackle climate change and sustainable development. *Entrepreneurship Theory and Practice, 45*(5), 999–1027.

Ghobakhloo, M. (2020). Industry 4.0, digitization, and opportunities for sustainability. *Journal of Cleaner Production, 252*, Article 119869.

Ghobakhloo, M., Asadi, S., Iranmanesh, M., Foroughi, B., Mubarak, M. F., & Yadegaridehkordi, E. (2023). Intelligent automation implementation and corporate sustainability performance: The enabling role of corporate social responsibility strategy. *Technology in Society, 74*, Article 102301.

Gill, S. S., Wu, H., Patros, P., Ottaviani, C., Arora, P., Pujol, V. C., Haunschild, D., Parlikad, A. K., Cetinkaya, O., & Lutfiyya, H. (2024). Modern computing: Vision and challenges. *Telematics and Informatics Reports*, Article 100116.

Gopal, S., & Pitts, J. (2025). GenAI: Unlocking sustainability insights and driving change in fintech. In *The FinTech revolution: Bridging geospatial data science, AI, and sustainability* (pp. 345–393). Springer.

Grybauskas, A., Stefanini, A., & Ghobakhloo, M. (2022). Social sustainability in the age of digitalization: A systematic literature review on the social implications of industry 4.0. *Technology in Society, 70*, Article 101997.

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2013). *Multivariate data analysis: Pearson new international edition PDF eBook*. Pearson Higher Ed.

Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019). When to use and how to report the results of PLS-SEM. *European Business Review, 31*(1), 2–24.

Hair, J. F., Sarstedt, M., Pieper, T. M., & Ringle, C. M. (2012). The use of partial least squares structural equation modeling in strategic management research: A review of past practices and recommendations for future applications. *Long Range Planning, 45*(5–6), 320–340.

Hendriarto, P. (2021). Understanding of the role of digitalization to business model and innovation: Economics and business review studies. *Linguistics and Culture Review, 5*(S1), 160–173.

Herden, C. J., Alliu, E., Cakici, A., Cormier, T., Deguelle, C., Gambhir, S., Griffiths, C., Gupta, S., Kamani, S. R., Kiratli, Y.-S., Kispataki, M., Lange, G., Moles de Matos, L., Tripero Moreno, L., Betancourt Nunez, H. A., Pilla, V., Raj, B., Roe, J., Skoda, M., & Edinger-Schons, L. M. (2021). Corporate digital responsibility. *Sustainability Management Forum | NachhaltigkeitsManagementForum, 29*(1), 13–29. https://doi.org/10.1007/s00550-020-00509-x

Hsu, C.-C., & Sandford, B. A. (2007). The Delphi technique: Making sense of consensus. *Practical Assessment, Research and Evaluation, 12*(1).

Jones, P., & Wynn, M. (2021). The leading digital technology companies and their approach to sustainable development. *Sustainability, 13*(12), 6612.

Keeney, S., Hasson, F., & McKenna, H. (2006). Consulting the oracle: Ten lessons from using the Delphi technique in nursing research. *Journal of Advanced Nursing, 53*(2), 205–212.

Kruk, S. R., Toonen, H. M., & Bush, S. R. (2023). Digital sustainability assurance governing global value chains: The case of aquaculture. *Regulation & Governance*.

Kunz, W. H., & Wirtz, J. (2024). Corporate digital responsibility (CDR) in the age of AI: Implications for interactive marketing. *The Journal of Research in Indian Medicine, 18*(1), 31–37.

Kunz, W., Wirtz, J., Hartley, N., & Tarbit, J. (2024). The importance of corporate digital responsibility in a digital service world. In *The impact of digitalization on current marketing strategies* (pp. 183–193). Emerald Publishing Limited.

Lambrecht, A., & Tucker, C. (2019). Algorithmic bias? An empirical study of apparent gender-based discrimination in the display of STEM career ads. *Management Science, 65*(7), 2966–2981.

Lin, B., & Zhang, Q. (2023). Corporate environmental responsibility in polluting firms: Does digital transformation matter? *Corporate Social Responsibility and Environmental Management, 30*(5), 2234–2246.

Linkov, I., Trump, B. D., Poinsatte-Jones, K., & Florin, M.-V. (2018). Governance strategies for a sustainable digital world. *Sustainability, 10*(2), 440.

Lobschat, L., Mueller, B., Eggers, F., Brandimarte, L., Diefenbach, S., Kroschke, M., & Wirtz, J. (2021). Corporate digital responsibility. *Journal of Business Research, 122*, 875–888. https://doi.org/10.1016/j.jbusres.2019.10.006

MacKenzie, S. B., & Podsakoff, P. M. (2012). Common method bias in marketing: Causes, mechanisms, and procedural remedies. *Journal of Retailing, 88*(4), 542–555.

Manasakis, C. (2018). Business ethics and corporate social responsibility. *Managerial and Decision Economics, 39*(4), 486–497.

Martin, U. M., Thapa, U., & Aguinis, H. (2024). Punishing the good? How to minimize an unfair CSR-washing label. *Business Horizons, 67*(2), 199–207.

Memon, K. R., & Ooi, S. K. (2023). Identifying digital leadership's role in fostering competitive advantage through responsible innovation: A SEM-neural network approach. *Technology in Society, 75*, Article 102399.

Memon, M., Thurasamy, R., Ting, H., & Cheah, J. (2024). Purposive sampling: A review and guidelines for quantitative research. *Journal of Applied Structural Equation Modeling, 9*(1), 1–23.

Mihale-Wilson, C., Hinz, O., van der Aalst, W., & Weinhardt, C. (2022). Corporate digital responsibility. *Business & Information Systems Engineering, 64*(2), 127–132. https://doi.org/10.1007/s12599-022-00746-y

Mueller, B. (2022). Corporate digital responsibility. *Business & Information Systems Engineering, 64*(5), 689–700. https://doi.org/10.1007/s12599-022-00760-0

Nguyen, T., Pham, T., Le, Q., & Bui, T. (2020). Impact of corporate social responsibility on organizational commitment through organizational trust and organizational identification. *Management Science Letters, 10*(14), 3453–3462.

Nietsch, M. (2018). Corporate illegal conduct and directors' liability: An approach to personal accountability for violations of corporate legal compliance. *Journal of Corporate Law Studies, 18*(1), 151–184.

Ntim, C. G., & Soobaroyen, T. (2013). Corporate governance and performance in socially responsible corporations: New empirical insights from a Neo-Institutional framework. *Corporate Governance: An International Review, 21*(5), 468–494.

Okoli, C., & Pawlowski, S. D. (2004). The Delphi method as a research tool: An example, design considerations and applications. *Information and Management, 42*(1), 15–29.

Pan, S. L., & Nishant, R. (2023). Artificial intelligence for digital sustainability: An insight into domain-specific research and future directions. *International Journal of Information Management, 72*, Article 102668.

Pope, S., & Wæraas, A. (2016). CSR-washing is rare: A conceptual framework, literature review, and critique. *Journal of Business Ethics, 137*, 173–193.

Ritter, T., & Pedersen, C. L. (2020). Digitization capability and the digitalization of business models in business-to-business firms: Past, present, and future. *Industrial Marketing Management, 86*, 180–190.

Roša, A., & Lobanova, L. (2022). Ethical responsibility of a company in the context of digital transformation of work: Conceptual model. *Sustainability, 14*(18), Article 11365. https://www.mdpi.com/2071-1050/14/18/11365.

Royakkers, L., Timmer, J., Kool, L., & Van Est, R. (2018). Societal and ethical issues of digitization. *Ethics and Information Technology, 20*, 127–142.

Song, D., Tan, Z., Wang, W., & Zhai, R.-X. (2025). Digital transformation and corporate social responsibility engagement: Evidence from China. *International Review of Financial Analysis, 97*, Article 103805.

Sun, Z., Wang, W., Wang, W., & Sun, X. (2024). How does digital transformation affect corporate social responsibility performance? From the dual perspective of internal drive and external governance. *Corporate Social Responsibility and Environmental Management, 31*(2), 1156–1176.

Švarc, J., Dabić, M., & Tekavčič, M. (2024). Controversies surrounding digitalization: Discussions in four key domains. In *Digital entrepreneurship in science, technology and innovation* (pp. 353–372). Edward Elgar Publishing.

Tosza, S. (2021). Internet service providers as law enforcers and adjudicators. A public role of private actors. *Computer Law & Security Report, 43*, Article 105614.

Tóth, Z., & Blut, M. (2024). Ethical compass: The need for corporate digital responsibility in the use of artificial intelligence in financial services. *Organizational Dynamics*, Article 101041.

Tran, T., Trinh, T.-P.-T., Le, C.-M., Hoang, L.-K., & Pham, H.-H. (2020). Research as a base for sustainable development of universities: Using the Delphi method to explore factors affecting international publishing among Vietnamese academic staff. *Sustainability, 12*(8), 3449.

Trittin-Ulbrich, H., & Böckel, A. (2022). Institutional entrepreneurship for responsible digital innovation: The case of corporate digital responsibility. *Creativity and Innovation Management, 31*(3), 447–459.

Ullah, A., Dogan, M., Pervaiz, A., Bukhari, A. A. A., Akkus, H. T., & Dogan, H. (2024). The impact of digitalization, technological and financial innovation on environmental quality in OECD countries: Investigation of N-shaped EKC hypothesis. *Technology in Society*, Article 102484.

Voigt, P., & Von dem Bussche, A. (2017). The eu general data protection regulation (gdpr). In *A practical guide* (1st ed.). Cham: Springer International Publishing, 10 (3152676), 10-5555.

Weber-Lewerenz, B. (2021). Corporate digital responsibility (CDR) in construction engineering—ethical guidelines for the application of digital transformation and artificial intelligence (AI) in user practice. *SN Applied Sciences, 3*, 1–25.

Wirtz, J., Kunz, W. H., Hartley, N., & Tarbit, J. (2023). Corporate digital responsibility in service firms and their ecosystems. *Journal of Service Research, 26*(2), 173–190.

Wut, T. M., Lee, D., Ip, W. M., & Lee, S. W. (2021). Digital sustainability in the organization: Scale development and validation. *Sustainability, 13*(6), 3530.